

AC Motor 2190 DAYS/TA PM CHECKLIST

For critical 480V motor

EQ#: _____ Unit: _____ SN# _____ WO# _____ Date: _____ PM Cycle: 2190 Days or TA

EQ	PM Item	Standard	Result	Comment						
Motor	1 Winding Insulation Resistance Measurement	Insulation resistance test 480V Motor: Megger 1.5MΩ (1KV test 60s)	PASS ☐ Abnormal ☐ Phase-Ground (MΩ): <table border="1"> <tr> <td>A-G</td> <td>B-G</td> <td>C-G</td> </tr> <tr> <td> </td> <td> </td> <td> </td> </tr> </table>	A-G	B-G	C-G				
	A-G	B-G	C-G							
	2 Dielectric Absorption Ratio, Polarization Index (P.I.) Measurement	Absorption ratio R60/R30 > 1.25 Polarization Index P.I. = R10/R1 > 2	PASS ☐ Abnormal ☐							
	3 Three-Phase Winding Resistance Measurement	Phase resistance deviation within ±3%	PASS ☐ Abnormal ☐ Phase-Phase (Ω): <table border="1"> <tr> <td>A-B</td> <td>B-C</td> <td>C-A</td> </tr> <tr> <td> </td> <td> </td> <td> </td> </tr> </table>	A-B	B-C	C-A				
	A-B	B-C	C-A							
	4 Appearance	No rust, no damage, no contamination; nameplate legible	PASS ☐ Abnormal ☐							
	5 Grounding	Grounding conductor (motor frame) firmly connected; no looseness; grounding resistance < 5 Ω	PASS ☐ Abnormal ☐ _____ Ω							
	6 Accessories	No abnormalities; indicators function properly	PASS ☐ Abnormal ☐							
	7 Terminal Box & Conduit	Connection secure; conduit in good condition; no water	PASS ☐ Abnormal ☐							
	8 Terminal Box Internal Wiring (From motor)	No discoloration or deterioration	PASS ☐ Abnormal ☐							
	9 Terminal Box Terminal Conductors (From outside)	No discoloration or deterioration; fixed securely; no stress damage	PASS ☐ Abnormal ☐							
10 Terminal Box Internal Terminal	Firmly fixed; no abnormalities	PASS ☐ Abnormal ☐								
11 Terminal Box Internal Terminal Insulation	Firmly secured; no looseness or damage	PASS ☐ Abnormal ☐								
12 Terminal Box Internal Ground Wires	Insulation intact; no looseness; grounding normal	PASS ☐ Abnormal ☐								
13 Terminal Box Internal Environment	Dry and clean; no moisture	PASS ☐ Abnormal ☐								

	14	PDMA Testing	Perform PDMA testing	PASS ☐ Abnormal ☐ N/A ☐	
Lubrication System	1	Automatic Grease Feeder	Lubrication piping unobstructed	PASS ☐ Abnormal ☐	
	2	Grease Refill	New grease must reach outlet after refill	PASS ☐ Abnormal ☐	
	3	Lubricating Oil Replacement	Oil level at half gauge (1/2)	PASS ☐ Abnormal ☐	
Cooling System	1	Cooling Fan	Fan grille clean and unobstructed	PASS ☐ Abnormal ☐	
	2	Cooling System Check	No rust; heat exchanger and duct clean and undamaged	PASS ☐ Abnormal ☒	
MCC or Starter	1	Breaker	Breaker Rating	_____ A	
			Current Limiter	_____ A	
			Trip Unit	PASS ☐ Abnormal ☐ N/A ☐	
			Current Limiter	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐	
	2	Contactor	Size	_____	
			Type	VFD ☐ R ☐ 1S ☐ 2S ☐	
			Main Contact	PASS ☐ Abnormal ☐ N/A ☐	
			Coil / Core	PASS ☐ Abnormal ☐ N/A ☐	
			Arc Chute	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐	
	3	Overload Relay	Heater Size	_____	
			Heaters	PASS ☐ Abnormal ☐ N/A ☐	
			Reset Mechanism	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐	
	4	Others	Protection Relay	PASS ☐ Abnormal ☐ N/A ☐	
			Auxiliary Relay	PASS ☐ Abnormal ☐ N/A ☐	
			CPT	PASS ☐ Abnormal ☐ N/A ☐	
			CT	PASS ☐ Abnormal ☐ N/A ☐	
			Fuses	PASS ☐ Abnormal ☐ N/A ☐	
			Push Buttons	PASS ☐ Abnormal ☐ N/A ☐	
			Indicating Lights	PASS ☐ Abnormal ☐ N/A ☐	

			Terminal Blocks	PASS ☐ Abnormal ☐ N/A ☐	
			Control Wiring	PASS ☐ Abnormal ☐ N/A ☐	
			Wire Raceway	PASS ☐ Abnormal ☐ N/A ☐	
			Handle Mechanism	PASS ☐ Abnormal ☐ N/A ☐	
			Discoloration	PASS ☐ Abnormal ☐ N/A ☐	
			Soft-Start	PASS ☐ Abnormal ☐ N/A ☐	
			Mechanical Interlock	PASS ☐ Abnormal ☐ N/A ☐	
			Auto Restart Function Check	PASS ☐ Abnormal ☐ N/A ☐	
			Motor Starter Cubicle Cleanness	PASS ☐ Abnormal ☐ N/A ☐	
Complete PM	1	Complete work	Close the cabinet and door	Yes ☐ No ☐	
	2	THM check	Clean the environment when complete the PM	Yes ☐ No ☐	

Rev. 0

Manager:_____ **Engineer:**_____ **Review By:**_____ **Inspect By:**_____

AC Motor 2190 DAYS/TA PM CHECKLIST

For critical 5kV&15kV motor

EQ#: _____ Unit: _____ SN# _____ WO# _____ Date: _____ PM Cycle: 2190 Days or TA

EQ	PM Item	Standard	Result	Comment						
Motor	1 Winding Insulation Resistance Measurement	Insulation resistance test 4.16kV Motor: Megger 5MΩ (2.5KV test 60s) 13.8kV Motor: Megger 12MΩ (5KV test 60s)	PASS ☐ Abnormal ☐ Phase-Ground (MΩ): <table border="1"> <tr> <td>A-G</td> <td>B-G</td> <td>C-G</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </table>	A-G	B-G	C-G				
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	6 Accessories	No abnormalities; indicators function properly	PASS ☐ Abnormal ☐							
	7 Terminal Box & Conduit	Connection secure; conduit in good condition; no water ingress	PASS ☐ Abnormal ☐							
	8 Terminal Box Internal Wiring (From motor)	No discoloration or deterioration	PASS ☐ Abnormal ☐							
	9 Terminal Box Terminal Conductors (From outside)	No discoloration or deterioration; fixed securely; no stress damage	PASS ☐ Abnormal ☐							
10 Terminal Box Internal Terminal	Firmly fixed; no abnormalities	PASS ☐ Abnormal ☐								
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	14	PDMA Testing	Perform PDMA testing	PASS ☐ Abnormal ☐ N/A ☐					
Lubrication System	1	Automatic Grease Feeder	Lubrication piping unobstructed	PASS ☐ Abnormal ☐					
	2	Grease Refill	New grease must reach outlet after refill	PASS ☐ Abnormal ☐					
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Cooling System	1	Cooling Fan	Fan grille clean and unobstructed	PASS ☐ Abnormal ☐					
	2	Cooling System Check	No rust; heat exchanger and duct clean and undamaged	PASS ☐ Abnormal ☒					
MCC or Starter	1	Breaker	SWGR No#	_____					
			Breaker Rating	_____ A					
			Setting	_____ A					
			Running Amps	<table border="1"> <tr> <td>A</td><td>B</td><td>C</td></tr> <tr> <td></td><td></td><td></td></tr> </table>	A	B	C		
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	2	MCC/Starter Inspection	Power Fuse	PASS ☐ Abnormal ☐ N/A ☐					
			CPT Fuse	PASS ☐ Abnormal ☐ N/A ☐					
			Insulators	PASS ☐ Abnormal ☐ N/A ☐					
			Phase Barriers	PASS ☐ Abnormal ☐ N/A ☐					
			Vacuum Interrupters	PASS ☐ Abnormal ☐ N/A ☐					
			Magnetic Coil	PASS ☐ Abnormal ☐ N/A ☐					
			Protection Relays	PASS ☐ Abnormal ☐ N/A ☐					
			Control Transformer	PASS ☐ Abnormal ☐ N/A ☐					
			Wire Connection	PASS ☐ Abnormal ☐ N/A ☐					
			Push Buttons	PASS ☐ Abnormal ☐ N/A ☐					
			Indication Lights	PASS ☐ Abnormal ☐ N/A ☐					
			Mechanical Interlock	PASS ☐ Abnormal ☐ N/A ☐					
			VFD	PASS ☐ Abnormal ☐ N/A ☐					
Complete PM			1	Complete work	Close the cabinet and door	Yes ☐ No ☐			
			2	THM check	Clean the environment when complete the PM	Yes ☐ No ☐			